



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year:2025), B.Sc. in CSE (Day)

Lab task-2
Course Title: Artificial Intelligence Lab
Course Code: CSE-316 Section: 222_D2

Student Details

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<u>Task-2 Status</u>	
Marks:	Signature:.....
Comments:.....	Date:.....

Question 1: How to reshape an array using numpy?

Sol:

main.py		Output
<pre>1 import numpy as np 2 arr = np.array([17, 42, 43, 22, 65, 65, 37, 84]) 3 zehad = arr.reshape(4, 2) 4 print(zehad) 5</pre>	Run	<pre>[[17 42] [43 22] [65 65] [37 84]]</pre>

Question 2: How to get the common items between two python numpy arrays?

Sol:

main.py		Output
<pre>1 #Zehad_222902069 2 import numpy as np 3 arr_store1 = np.array([174, 237, 36, 400, 502]) 4 arr_store2 = np.array([502, 574, 69, 36, 80]) 5 common_items = np.intersect1d(arr_store1, arr_store2) 6 print("Common items=", common_items)</pre>	Run	<pre>Common items= [36 502] === Code Execution Successful ===</pre>

Question 3: How to get the common items as Transpose and Flatten of a numpy array?

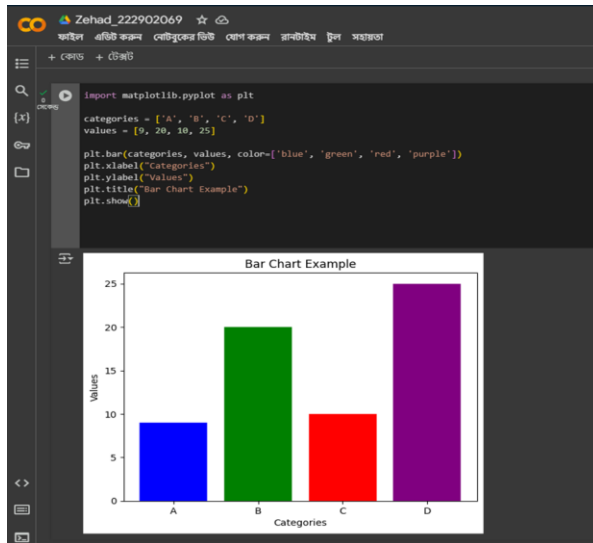
Sol:

main.py		Output
<pre>1 #Zehad_222902069 2 3 import numpy as np 4 arr_num1 = np.array([[176, 32, 36], [43, 54, 17]]) 5 arr_num2 = np.array([[34, 32, 61], [17, 32, 69]]) 6 7 common_items = np.intersect1d(arr_num1, arr_num2) 8 print("Common Items=", common_items, "\n") 9 10 transposed = arr_num1.T 11 print("Transposed Array=\n", transposed, "\n") 12 13 flattened = arr_num1.flatten() 14 print("Flattened Array=", flattened, "\n")</pre>	Run	<pre>Common Items= [17 32] Transposed Array= [[176 43] [32 54] [36 17]] Flattened Array= [176 32 36 43 54 17] === Code Execution Successful ===</pre>

Question 4: Draw a bar chart and pie chart using matplotlib?

Sol:

1. Bar Chart:



2. Pie Chart:

